Pacha Hruthik

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.97
VISWA SAI JUNIOR COLLEGE	Class XII (BOARD OF INTERMEDIATE EDUCATION ANDHARA PRADESH)	2023	92.20%
SRI SRINIVASA EM HIGH SCHOOL	Class X (BOARD OF SECONDARY EDUCATION ANDHARA PRADESH)	2021	100.00%

PROJECTS

Multi-Client Quiz System | C++, TCP/IP, Multithreading

Course Project under Prof. Kotaro Kataoka and Prof. Praveen Tammana

- Developed a multithreaded **TCP client-server quiz system** supporting concurrent clients, real-time scoring, leaderboards, and session management.
- Integrated an **LLM-based question generator** to dynamically create multiple-choice quizzes based on user-selected genres.
- Implemented **thread-safe server state and keep-alive mechanisms** using mutexes, atomic variables, and custom message protocols.
- Added **server-side caching with expiration** to reuse generated questions and reduce repeated LLM requests; built a Python-based automated multi-client tester.

Cricket Tournament Management System | React, FastAPI, MySQL

Course Project under Prof. Shirshendu Das

- Built a full-stack cricket tournament management application to manage **tournaments, teams, players, matches, live scores, and points tables**.
- Developed modular **FastAPI REST APIs** backed by MySQL for authentication, tournament management, team/player operations, and match tracking.
- Implemented **JWT authentication, bcrypt password hashing, and role-based authorization** for managers, organizers, administrators, and viewers.
- Developed ball-by-ball match logging and automated calculation of **player statistics, team scores, wickets, match results, and net run rate**.

Infinite Precision Arithmetic | C++,OOP

Course Project under Prof. Ramakrishna Upadrasta and Prof. Rajesh kedia

- Developed an **arbitrary-precision arithmetic library in C++** to process integers and floating-point numbers exceeding native data-type limits.
- Implemented **addition, subtraction, multiplication, and division** using string-based arithmetic algorithms for large signed integers.
- Implemented **floating-point arithmetic with decimal-position handling and sign management**, supporting high-precision calculations.
- Used **operator overloading and object-oriented design** to create reusable Integer and Float classes, with a CMake-built command-line interface.

SKILLS

Languages: C,C++,Python,SQL

Machine Learning: Deep Learning,Transformers,LLMs

RELEVANT COURSES

Core Courses: Data Structures, Algorithms, Operating Systems, Computer Architecture, Database Management Systems,Computer Networks,Software Engineering

AI/ML: Machine Learning, Multi Agent Systems

MATHS: Linear Optimization,Theory Of Computation, Probability and Random Variables

POSITIONS OF RESPONSIBILITY

Transport Coordinator -- Coordinated student transportation, bus schedules, and logistics to ensure smooth and timely operations during college activities and events.

Medical & Security Coordinator, **Inter-IIT Sports Meet** -- Assisted with participant safety, basic medical support, and security coordination to ensure smooth conduct of the event.